

Answers to Sample Questions
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1. Indicate whether each of the following statements is TRUE or FALSE.
 - a. Answer: False
 - b. Answer: True
 - c. Answer: True
 - d. Answer: False
 - e. Answer: True
 - f. Answer: False
 - g. Answer: False
 - h. Answer: True
 - i. Answer: True

2. Multiple choice questions. Only one answer is correct for each question. Circle the right choice.
 - a. Answer: 3)
 - b. Answer: 2)
 - c. Answer: 3)
 - d. Answer: 2)
 - e. Answer: 4)
 - f. Answer: 2)
 - g. Answer: 3)
 - h. Answer: 4)
 - i. Answer: 4)
 - j. Answer: 4)

3. Short-Answer Questions

a. Answer:

Advantages:

- The greatest possible amount of data is used for training in each iteration.
- Deterministic: no random sampling is involved

Disadvantages:

- Long running time
- Non-stratified partitions of the data set

b. Answer:

Neural network-based text embedding methods offer several advantages over traditional bag-of-words (BoW) representations. The major ones are:

- Neural embeddings capture semantic meaning, allowing similar words (e.g., "cat" and "kitten") to have similar vector representations. This enables tasks like analogy resolution (e.g., "king" - "man" + "woman" $\approx$ "queen").
- Compact Dense Vectors: Embeddings use fixed-size, dense vectors (e.g., 300–768 dimensions), reducing dimensionality compared to BoW's sparse, vocabulary-sized vectors. This mitigates the "curse of dimensionality" and improves computational efficiency.

c. Answer:

Class +

d. Answer:

It helps determine if observed performance differences are due to chance. Using p-values from tests like paired t-test, we assess if the null hypothesis (no difference) can be rejected under assumptions like normality.

e. Answer: To initialize weights in a way that captures useful features before training or fine-tuning.

f. Answer:

Accuracy = $(80+90)/200 = 85\%$

Precision = $80 / (80+10) = 88.89\%$

Recall = $80 / (80+20) = 80\%$

F1-score $\approx 84.21\%$

4. Answer:

Join step result:

Candidates: {a, b, c, d}, {a, b, c, e}, {a, b, d, e}, {a, c, d, e}, {b, c, d, e}

Prune step result:

{a, b, c, d}, {a, b, c, e}

5. Answer:

(1) Compute the prior probabilities:

$$P(\text{Yes}) = 6/10 = 0.6$$

$$P(\text{No}) = 4/10 = 0.4$$

(2) Compute conditional probabilities:

$$P(\text{Age} = \text{Young} \mid \text{Buy} = \text{Yes}) = 1/6$$

$$P(\text{Income} = \text{Medium} \mid \text{Buy} = \text{Yes}) = 2/6 = 1/3$$

$$P(\text{Age} = \text{Young} \mid \text{Buy} = \text{No}) = 3/4$$

$$P(\text{Income} = \text{Medium} \mid \text{Buy} = \text{No}) = 1/4$$

(3) Classify the test example:

Age = Young, Income = Medium

For Buy = Yes:

$$P(\text{Buy} = \text{Yes}) \cdot P(\text{Age} = \text{Young}, \text{Income} = \text{Medium} \mid \text{Buy} = \text{Yes})$$

$$= P(\text{Buy} = \text{Yes}) \cdot P(\text{Age} = \text{Young} \mid \text{Buy} = \text{Yes}) \cdot P(\text{Income} = \text{Medium} \mid \text{Buy} = \text{Yes})$$

$$= 0.6 \cdot \frac{1}{6} \cdot \frac{1}{3} = 0.0333$$

For Buy = No:

$$P(\text{Buy} = \text{No}) \cdot P(\text{Age} = \text{Young}, \text{Income} = \text{Medium} \mid \text{Buy} = \text{No})$$

$$= P(\text{Buy} = \text{No}) \cdot P(\text{Age} = \text{Young} \mid \text{Buy} = \text{No}) \cdot P(\text{Income} = \text{Medium} \mid \text{Buy} = \text{No})$$

$$= 0.4 \cdot \frac{3}{4} \cdot \frac{1}{4} = 0.075$$

Compare:

- $P(\text{Buy} = \text{Yes} \mid \text{Age} = \text{Young}, \text{Income} = \text{Medium}) \propto 0.033$
- $P(\text{Buy} = \text{No} \mid \text{Age} = \text{Young}, \text{Income} = \text{Medium}) \propto 0.075$

Thus, the prediction is Buy=No.

6. Answer:

Assume that each variable has values of 0 and 1. The probabilities needed to complete the belief network are the ones in the conditional probability table of each variable.

$P(S=0)$, $P(S=1)$ (Listing one of them is ok since they sum to 1).

$P(L=0 \mid S=0)$, $P(L=1 \mid S=0)$, $P(L=0 \mid S=1)$, $P(L=1 \mid S=1)$ (Listing two of them is ok as long as the remaining ones can be derived from the listed ones. For example, $P(L=1 \mid S=0) = 1 - P(L=0 \mid S=0)$)

$P(C=0 \mid S=0)$, $P(C=1 \mid S=0)$, $P(C=0 \mid S=1)$, $P(C=1 \mid S=1)$ (Same as the above)

$P(F=0|L=0,C=0)$, $P(F=1|L=0,C=0)$, $P(F=0|L=0,C=1)$, $P(F=1|L=0,C=1)$, $P(F=0|L=1,C=0)$, $P(F=1|L=1,C=0)$, $P(F=0|L=1,C=1)$, $P(F=1|L=1,C=1)$ (Similarly, listing 4 of these is ok as long as the remaining ones can be derived from the listed ones).

Thus, the number of probabilities is 18 (9 is considered to be right as well since the other 9 can be derived).

7. Answer;

(1) Network Structure

- **Input layer:** 2 inputs
- **Output layer:** 1 neuron
- **No hidden layers**

(2) Activation Function

- **Step function** (i.e., output is 1 if input ≥ 0 , otherwise 0)

(3) Weights and Bias

Let the weights be:

$$w_1=1, w_2=1, b=-1.5$$

The neuron computes:

$$z=w_1x_1+w_2x_2+b=x_1+x_2-1.5$$

Apply the step function:

- If $x_1=x_2=1$: $z = 0.5 \geq 0 \rightarrow \text{output} = 1$
- Else: $z < 0 \rightarrow \text{output} = 0$

(4) Explanation

Only when both inputs are 1 does the sum reach or exceed the threshold for activation. This correctly implements the AND function.

x1 x2 Output

0	0	0
0	1	0
1	0	0
1	1	1